

Example 1 : A survey of 500 from a school produced the following information. 200 play volleyball, 120 play hockey. 60 play both volleyball and hockey. How many are not playing either volleyball or hockey?

Solution :

Let A denote the students who play Volleyball.

Let B denote the students who play Hockey.

It is given that

$$n = 500$$

$$|A| = 200$$

$$|B| = 120$$

$$|A \cap B| = 60$$

By principle of Inclusion – Exclusion, the number of students playing either Volleyball or Hockey is

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B| = 200 + 120 - 60$$

$$= 260$$

$$\therefore \left. \begin{array}{l} \text{The number of students} \\ \text{not playing either Volleyball} \\ \text{or Hockey is} \end{array} \right\} = 500 - 260$$

$$= 240$$

Example 2 : A total of 1232 students have taken a course in Tamil, 879 have taken a course in Telugu, and 114 have taken a course in Hindi. Further 103 have taken a course in both Tamil and Telugu, 23 have taken a course in Tamil and Hindi, and 14 have taken a course in Telugu and Hindi. If 2092 students have taken atleast one of the Tamil, Telugu and Hindi, how many students have taken a course in all three languages.

Solution :

Let A be the students who have taken a course in Tamil.

Let B be the students who have taken a course in Telugu.

Let C be the students who have taken a course in Hindi.

$$\text{Then, } |A| = 1232; \quad |B| = 879; \quad |C| = 114;$$

$$|A \cap B| = 103; \quad |A \cap C| = 23; \quad |B \cap C| = 14;$$

$$\text{and } |A \cup B \cup C| = 2092$$

By principle of Inclusion – Exclusion, we have

$$\begin{aligned}
 |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\
 &\quad - |B \cap C| + |A \cap B \cap C| \\
 2092 &= 1232 + 879 + 114 - 103 - 23 - 14 \\
 &\quad + |A \cap B \cap C|
 \end{aligned}$$

$$|A \cap B \cap C| = 2232 - 2225 = 7$$

Therefore, there are 7 students who have taken the course in Tamil, Telugu and Hindi.

Example 3 : In a survey of 100 students, it was found that 30 studied Mathematics, 54 studied Statistics, 25 studied Operation Research, 1 studied all the three subjects. 20 studied Mathematics and Statistics, 3 studied Mathematics and Operations Research and 15 studied Statistics and Operations Research.

(i) How many students studied none of these subjects?

(ii) How many students studied only Mathematics?

Solution :

(i) Let A denote the students who studied Mathematics.

Let B denote the students who studied Statistics.

Let C denote the students who studied Operations Research.

Then, $|A| = 30$; $|B| = 54$; $|C| = 25$;

$|A \cap B| = 20$; $|A \cap C| = 3$; $|B \cap C| = 15$;

and $|A \cap B \cap C| = 1$

By principle of Inclusion – Exclusion, students who studied any one of the subject is

$$\begin{aligned}
 |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\
 &\quad - |B \cap C| + |A \cap B \cap C| \\
 &= 30 + 54 + 25 - 20 - 3 - 15 + 1 \\
 &= 110 - 38 = 72
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Students who studied none} \\
 \text{of these 3 subjects} \} &= 100 - 72 \\
 &= 28
 \end{aligned}$$

(ii) Now,

$$\begin{aligned}
 \text{The number of students} \\
 \text{studied only Mathematics} \\
 \text{and Statistics} \} &= n(A \cap B) - n(A \cap B \cap C)
 \end{aligned}$$

$$= 20 - 1 = 19$$

$$\left. \begin{array}{l} \text{The number of students} \\ \text{studied only Mathematics} \\ \text{and Operations Research} \end{array} \right\} = n(A \cap C) - n(A \cap B \cap C)$$

$$= 3 - 1 = 2$$

Then,

$$\left. \begin{array}{l} \text{The number of students} \\ \text{studied only Mathematics} \end{array} \right\} = 30 - 19 - 2$$

$$= 9$$

Example 4 : How many positive integers not exceeding 1000 are divisible by 7 or 11?

Solution :

Let A denote the set of +ve integers not exceeding 1000 that are divisible by 7:

Let B denote the set of +ve integers not exceeding 1000 that are divisible by 11.

$$\text{Then, } |A| = \left[\frac{1000}{7} \right] = [142.8] = 142$$

$$|B| = \left[\frac{1000}{11} \right] = [90.9] = 90$$

$$|A \cap B| = \left[\frac{1000}{7 \times 11} \right] = [12.9] = 12$$

The number of positive integer not exceeding 1000 that are divisible either 7 or 11 is $|A \cup B|$

By Principle of Inclusion – Exclusion

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 142 + 90 - 12 \\ &= 220 \end{aligned}$$

There are 220 +ve integer not exceeding 1000 divisible by either 7 or 11.

Example 5 : Determine n such that $1 \leq n \leq 100$ which are not divisible by 5 or by 7.

Solution :

Let A denote the number n , $1 \leq n \leq 100$ which is divisible by 5.

Let B denote the number n , $1 \leq n \leq 100$ which is divisible by 7.

$$\text{Then, } |A| = \left[\frac{100}{5} \right] = [20] = 20$$

$$|B| = \left[\frac{100}{7} \right] = [14.3] = 14$$

$$|A \cap B| = \left[\frac{100}{5 \times 7} \right] = [2.8] = 2$$

Now, the number n , $1 \leq n \leq 100$ which is divisible by either 5 or 7 is $|A \cup B|$.

By Principle of Inclusion – Exclusion

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 20 + 14 - 2 \\ &= 32 \end{aligned}$$

The number n , $1 \leq n \leq 100$ which is not divisible by either 5 or 7 is

$$\begin{aligned} &= 100 - 32 \\ &= 68 \end{aligned}$$

There are 68 number not exceeding 100 that are not divisible by either 5 or 7.

Example 6 : A survey among 100 students shows that of the three ice cream flavours Vanilla, Chocolate and Strawberry, 50 students like Vanilla, 43 like Chocolate, 28 like Strawberry, 13 like Vanilla and Chocolate, 11 like Chocolate and Strawberry, 12 like Strawberry and Vanilla, and 5 like all of them. Find the number of students surveyed who like each of the following flavours.

- (i) Chocolate but not Strawberry.
- (ii) Chocolate and Strawberry, but not Vanilla.
- (iii) Vanilla or Chocolate, but not Strawberry.

Solution :

Let A denote the set of students who like Vanilla.

Let B denote the set of students who like Chocolate.

Let C denote the set of students who like Strawberry.

Since 5 students like all flavours, $|A \cap B \cap C| = 5$.

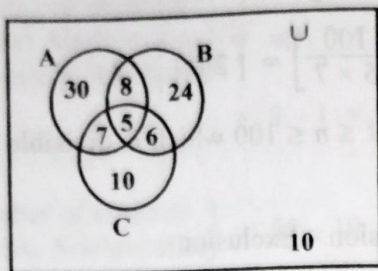
12 Students like both Strawberry and Vanilla.

$$\therefore |A \cap C| = 12$$

But 5 of them like Chocolate also. Therefore,

$$|(A \cap C) - B| = 7$$

Similarly, $|(B \cap C) - A| = 6$



Of the 28 students who like Strawberry, we have already accounted for

$$7 + 5 + 6 = 18$$

So, the remaining 10 students belong to the set $C - (A \cup B)$. Similarly,

$$|A - (B \cup C)| = 30 \text{ and } |B - (A \cup C)| = 24$$

Thus far, we have accounted for 90 of the 100 students. The remaining 10 students like outside the region $A \cup B \cup C$.

Now,

$$(i) \quad |B - C| = 24 + 8 = 32$$

So, 32 students like Chocolate but not Strawberry.

$$(ii) \quad |(B \cap C) - A| = 6$$

Therefore, 6 students like both Chocolate and Strawberry, but not Vanilla.

$$(iii) \quad |(A \cup B) - C| = 30 + 8 + 24 = 62$$

Therefore 62 students like Vanilla or Chocolate, but not Strawberry.

Example 7: Find the number of integers between 1 to 250 that are not divisible by any of the integers 2, 3, 5 and 7.

Solution :

Let A denote the integer from 1 to 250 that are divisible by 2.

Let B denote the integer from 1 to 250 that are divisible by 3.

Let C denote the integer from 1 to 250 that are divisible by 5.

Let D denote the integer from 1 to 250 that are divisible by 7.

$$|A| = \left\lfloor \frac{250}{2} \right\rfloor = 125$$

$$|B| = \left[\frac{250}{3} \right] = 83$$

$$|C| = \left[\frac{250}{5} \right] = 50$$

$$|D| = \left[\frac{250}{7} \right] = 35$$

Now,

$$\left. \begin{array}{l} \text{The number of integer} \\ \text{between 1 - 250 that} \\ \text{are divisible by 2 \& 3} \end{array} \right\} = |A \cap B| = \left[\frac{250}{2 \times 3} \right] = 41$$

$$\left. \begin{array}{l} \text{The number of integer} \\ \text{divisible by 2 \& 5} \end{array} \right\} = |A \cap C| = \left[\frac{250}{2 \times 5} \right] = 25$$

Similarly,

$$|A \cap D| = \left[\frac{250}{2 \times 7} \right] = 17$$

$$|B \cap C| = \left[\frac{250}{3 \times 5} \right] = 16$$

$$|B \cap D| = \left[\frac{250}{3 \times 7} \right] = 11$$

$$|C \cap D| = \left[\frac{250}{5 \times 7} \right] = 7$$

$$\left. \begin{array}{l} \text{The number of integer} \\ \text{divisible by 2, 3 \& 5} \end{array} \right\} = |A \cap B \cap C| = \left[\frac{250}{2 \times 3 \times 5} \right] = 8$$

Similarly

$$|A \cap B \cap D| = \left[\frac{250}{2 \times 3 \times 7} \right] = 5$$

$$|A \cap C \cap D| = \left[\frac{250}{2 \times 5 \times 7} \right] = 3$$

$$|B \cap C \cap D| = \left[\frac{250}{3 \times 5 \times 7} \right] = 2$$

$$|A \cap B \cap C \cap D| = \left[\frac{250}{2 \times 3 \times 5 \times 7} \right] = 1$$

The number of integers between 1 – 250 that are divisible by 2, 3, 5 and 7 is $|A \cup B \cup C \cup D|$.

By principle of Inclusion – Exclusion

$$\begin{aligned}
 |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| - |A \cap B| \\
 &\quad - |A \cap C| - |A \cap D| - |B \cap C| \\
 &\quad - |B \cap D| - |C \cap D| + |A \cap B \cap C| \\
 &\quad + |A \cap B \cap D| + |A \cap C \cap D| \\
 &\quad + |B \cap C \cap D| - |A \cap B \cap C \cap D| \\
 &= (125 + 83 + 50 + 35) \\
 &\quad - (41 + 25 + 17 + 16 + 11 + 7) \\
 &\quad + (8 + 5 + 3 + 2) - 1 \\
 &= 293 - 117 + 18 - 1 = 193
 \end{aligned}$$

Now,

$$\begin{aligned}
 \left. \begin{array}{l} \text{The number of integer} \\ \text{not divisible by any} \\ \text{of 2, 3, 5 and 7} \end{array} \right\} &= \text{Total} - |A \cup B \cup C \cup D| \\
 &= 250 - 193 \\
 &= 57
 \end{aligned}$$

Example 8 : A survey shows that 57% of Indians like coffee whereas 75% like Tea. What can you say about the percentage of Indians who like both coffee and Tea.

Solution :

Let A denote set of Indians who like coffee.

Let B denote set of Indians who like Tea.

Assume the total population is 100. Therefore,

$$|A| = 57$$

$$|B| = 75$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 57 + 75 - |A \cap B|$$

$$|A \cup B| = 132 - |A \cap B|$$

Since $|A \cup B| \leq 100$, it follows that $|A \cap B| > 32$... (1)

Since $A \cap B \subseteq A$ and $(A \cap B) \subseteq B$,

We have $|A \cap B| \leq |A|$ and $|A \cap B| \leq |B|$

$$|A \cap B| \leq 57 \text{ and } |A \cap B| \leq 72$$

$$|A \cap B| \leq 57$$

... (2)

From (1) & (2), the percentage of Indians who likes both coffee and tea lies between 32 and 57.

Example 9 : Out of 100 students in a college, 38 play tennis, 57 play cricket and 31 play hockey, 9 play cricket and hockey, 10 play hockey and tennis, 12 play tennis and cricket.

How many play (1) All three games.

(2) Just one game.

(3) Tennis and cricket but not hockey.

(Assume that each student plays atleast one game.)

Solution :

Let T, C and H denote the set of students playing Tennis, Cricket and Hockey respectively.

Given

$$|T| = 38, \quad |C| = 57, \quad |H| = 31$$

$$|T \cap C| = 12, \quad |T \cap H| = 10, \quad |C \cap H| = 9$$

$$|T \cup C \cup H| = 100$$

Now

$$\left. \begin{array}{l} \text{The number of integer} \\ \text{who play all three games} \end{array} \right\} = |T \cap C \cap H|$$

By principle of Inclusion and Exclusion

$$|T \cup C \cup H| = |T| + |C| + |H| - |T \cap C| - |T \cap H| - |C \cap H| + |T \cap C \cap H|$$

$$100 = 38 + 57 + 31 - (12 + 10 + 9)$$

$$+ |T \cap C \cap H|$$

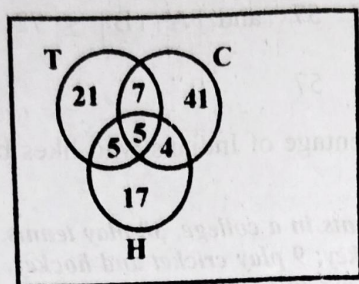
$$|T \cap C \cap H| = 100 - 126 + 31$$

$$= 131 - 126$$

$$= 5$$

$$\left. \begin{array}{l} \text{Number of students} \\ \text{who play all 3 games} \end{array} \right\} = 5$$

From the given data, we have



$$\begin{aligned}
 \text{Number of students playing just one game} &= \text{No. of students playing Tennis only} + \text{No. of students playing Cricket only} \\
 &\quad + \text{No. of students playing Hockey only} \\
 &= 21 + 41 + 17 = 79
 \end{aligned}$$

The number of students playing Tennis and Cricket but not Hockey.

$$\begin{aligned}
 &= |T \cap C| - |T \cap C \cap H| \\
 &= 12 - 5 = 7
 \end{aligned}$$

Example 10 : How many integers between 1 to 100 that are

- 1) not divisible by 7, 11, or 13
- 2) divisible by 3 but not by 7.

Solution :

- (1) Let A, B and C denote respectively the number of Integer between 1 to 100 that are divisible by 7, 11 and 13 respectively.

Now, $|A| = \left\lfloor \frac{100}{7} \right\rfloor = 14$

$$|B| = \left\lfloor \frac{100}{11} \right\rfloor = 9$$

$$|C| = \left\lfloor \frac{100}{13} \right\rfloor = 7$$

$$|A \cap B| = \left\lfloor \frac{100}{7 \times 11} \right\rfloor = 1$$

$$|A \cap C| = \left\lfloor \frac{100}{7 \times 13} \right\rfloor = 1$$

$$|B \cap C| = \left\lfloor \frac{100}{11 \times 13} \right\rfloor = 0$$

$$|A \cap B \cap C| = \left[\frac{100}{7 \times 11 \times 13} \right] = 0$$

The number of integers between 1 – 100 that are divisible by 7, 11, or 13 is $(A \cup B \cup C)$

By principle of Inclusion and exclusion

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\ &\quad - |B \cap C| + |A \cap B \cap C| \\ &= 14 + 9 + 7 - (1 + 1 + 0) + 0 \\ &= 30 - 2 \\ &= 28 \end{aligned}$$

Now,

$$\begin{aligned} \left. \begin{array}{l} \text{The number of integer} \\ \text{not divisible by any} \\ \text{of 7, 11, and 13} \end{array} \right\} &= \text{Total} - |A \cup B \cup C| \\ &= 100 - 28 \\ &= 72 \end{aligned}$$

- (2) Let A and B denote the no. between 1 – 100 that are divisible by 3 and 7 respectively.

$$|A| = \left[\frac{100}{3} \right] = 33$$

$$|B| = \left[\frac{100}{7} \right] = 14$$

$$|A \cap B| = \left[\frac{100}{3 \times 7} \right] = 4$$

The number of integers divisible by 3 but not by 7.

$$\begin{aligned} &= |A| - |A \cap B| \\ &= 33 - 4 \\ &= 29 \end{aligned}$$

Example 11 : Find the number of the integers between 1 to 100 that are divisible by (i) 2, 3, 5 or 7

(ii) 2, 3, 5 but not by 7.

Solution :

(i) Let A, B, C and D denote the number of positive integers between 1 – 100 that are divisible by 2, 3, 5 and 7 respectively.

$$\text{Now, } |A| = \left[\frac{100}{2} \right] = 50$$

$$|B| = \left\lfloor \frac{100}{3} \right\rfloor = 33$$

$$|C| = \left\lfloor \frac{100}{5} \right\rfloor = 20$$

$$|D| = \left\lfloor \frac{100}{7} \right\rfloor = 14$$

$$|A \cap B| = \left\lfloor \frac{100}{2 \times 3} \right\rfloor = 16$$

$$|A \cap C| = \left\lfloor \frac{100}{2 \times 5} \right\rfloor = 10$$

$$|A \cap D| = \left\lfloor \frac{100}{14} \right\rfloor = 7$$

$$|B \cap C| = \left\lfloor \frac{100}{3 \times 5} \right\rfloor = 6$$

$$|B \cap D| = \left\lfloor \frac{100}{3 \times 7} \right\rfloor = 4$$

$$|C \cap D| = \left\lfloor \frac{100}{5 \times 7} \right\rfloor = 2$$

$$|A \cap B \cap C| = \left\lfloor \frac{100}{2 \times 3 \times 5} \right\rfloor = 3$$

$$|A \cap B \cap D| = \left\lfloor \frac{100}{2 \times 3 \times 7} \right\rfloor = 2$$

$$|A \cap C \cap D| = \left\lfloor \frac{100}{2 \times 5 \times 7} \right\rfloor = 1$$

$$|B \cap C \cap D| = \left\lfloor \frac{100}{3 \times 5 \times 7} \right\rfloor = 0$$

$$|A \cap B \cap C \cap D| = 0$$

By the principle of Inclusion and Exclusion,

$$|A \cup B \cup C \cup D|$$

$$= |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |A \cap D|$$

$$- |B \cap C| - |B \cap D| + |A \cap B \cap C| + |A \cap B \cap D|$$

$$+ |A \cap C \cap D| + |B \cap C \cap D| - |A \cap B \cap C \cap D|$$

$$\begin{aligned}
 &= (50 + 33 + 20 + 14) \\
 &\quad - (16 + 10 + 7 + 6 + 4 + 2) \\
 &\quad + (3 + 2 + 1 + 0) - 0 \\
 &= 117 - 45 + 6 \\
 &= 123 - 45 \\
 &= 78
 \end{aligned}$$

(ii) The number of integers between 1 – 100 that are divisible 2, 3, 5 but not by 7

$$\begin{aligned}
 &= |A \cap B \cap C| - |A \cap B \cap C \cap D| \\
 &= 3 - 0 \\
 &= 3
 \end{aligned}$$

Example 12 : There are 2500 students in a college, of these 1700 have taken a course in C, 1000 have taken a course Pascal and 550 have taken a course in Networking. Further 750 have taken courses in both C and Pascal, 400 have taken courses in both C and Networking, and 275 have taken courses in both Pascal and Networking. If 200 of these students have taken courses in C, Pascal and Networking.

- (1) How many of these 2500 students have taken a course in any of these three courses C, Pascal and Networking?
- (2) How many of these 2500 students have not taken a course in any of these three courses C, Pascal and Networking?

Solution :

Let A, B and C denote students have taken a course in C, Pascal and Networking respectively.

$$\text{Given } |A| = 1700$$

$$|B| = 1000$$

$$|C| = 550$$

$$|A \cap B| = 750$$

$$|A \cap C| = 400$$

$$|B \cap C| = 275$$

$$|A \cap B \cap C| = 200$$

$$\left. \begin{array}{l} \text{Number of students who have taken} \\ \text{any one of these course} \end{array} \right\} = |A \cap B \cap C|$$

By principle of Inclusion and exclusion,

$$\begin{aligned}
 |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\
 &\quad - |B \cap C| + |A \cap B \cap C|
 \end{aligned}$$

$$\begin{aligned}
 &= (1700 + 1000 + 550) - (750 + 400 + 275) + 200 \\
 &= 3450 - 1425 \\
 &= 2025
 \end{aligned}$$

Number of students who have not taken any 3 courses

$$\begin{aligned}
 &= \text{Total} - |A \cup B \cup C| \\
 &= 2500 - 2025 \\
 &= 475
 \end{aligned}$$

Example 13 : Determine the number of positive integers n , $1 \leq n \leq 1000$, that are not divisible by 2, 3 or 5 but are divisible by 7.

Solution :

Let A, B, C and D denote respectively the number of integers between 1 – 1000 that are divisible by 2, 3, 5 and 7 respectively.

$$\begin{aligned}
 \text{Now } |D| &= \left\lfloor \frac{1000}{7} \right\rfloor \\
 &= [142.8] \\
 &= 142
 \end{aligned}$$

$$\begin{aligned}
 |A \cap B \cap C \cap D| &= \left\lfloor \frac{1000}{2 \times 3 \times 5 \times 7} \right\rfloor \\
 &= \left\lfloor \frac{1000}{210} \right\rfloor \\
 &= [4.7] \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 \left. \begin{array}{l} \text{The number between 1 – 1000} \\ \text{that are divisible by 7 but not} \\ \text{divisible by 2, 3, 5 and 7} \end{array} \right\} &= |D| - |A \cap B \cap C \cap D| \\
 &= 142 - 4 \\
 &= 138
 \end{aligned}$$

Example 14 : A survey of 500 television watchers produced the following information. 285 watch Hockey games. 195 watch Foot ball games. 115 watch basketball games. 70 watch football and hockey games. 50 watch hockey and basketball games and 30 watch football and hockey games. 50 do not watch any of the three games. How many people watch exactly one of the three games?

Solution :

H – Let Television watchers who watch Hockey.

F – Let Television watchers who watch Football.

B – Let Television watchers who watch Basket ball.

Given $n(H) = 285$

$n(F) = 195$

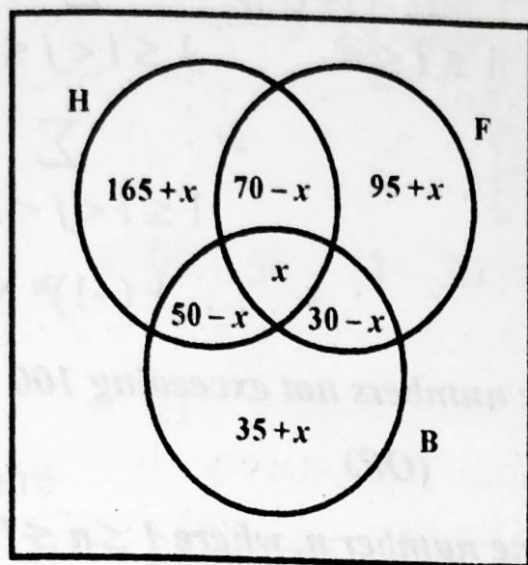
$n(B) = 115$

$n(H \cap F) = 70$

$n(H \cap B) = 50$

$n(F \cap B) = 30$

Let x be the number television watcher who watch all three games.
Now, we have



Given 50 members does not watch any of the three games.

Hence $(165 + x) + (95 + x) + (35 + x) + (70 - x)$

$+ (50 - x) + (30 - x) + x = 500$

$445 + x = 500$

$\Rightarrow x = 55$

Number of students who watches exactly one game is $\left. \begin{aligned} &= 165 + x + 95 + x + 35 + x \\ &= 295 + 3 \times 55 \\ &= 295 + 165 \\ &= 460 \end{aligned} \right\}$

$= 295 + 165$

$= 460$

$= 460$